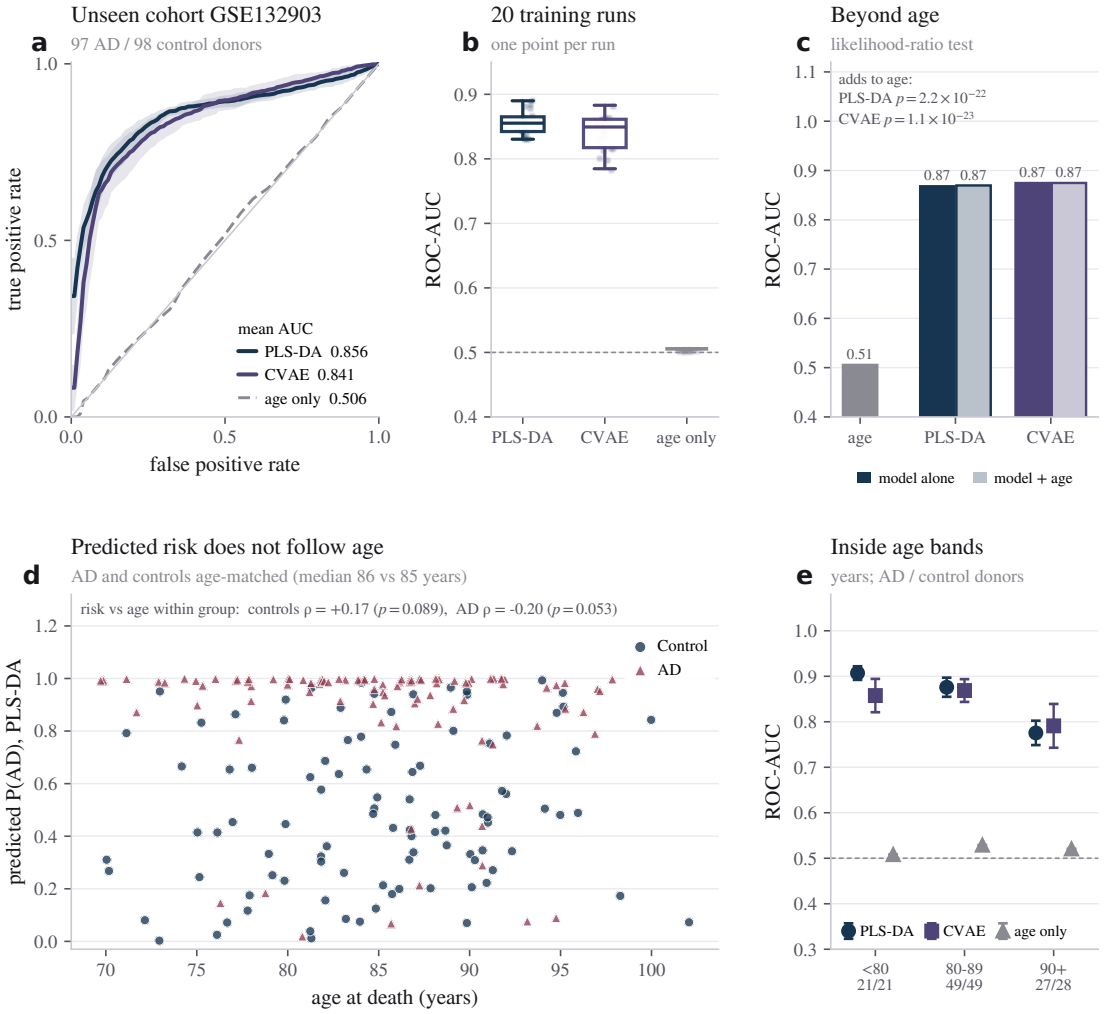
Unseen cohort GSE132903
a
97 AD / 98 control donors
true positive rate
false positive rate
mean AUC
PLS-DA 0.856
CVAE 0.841
age only 0.506
20 training runs
b
one point per run
ROC-AUC
PLS-DA
CVAE
age only
Beyond age
c
likelihood-ratio test
adds to age:
PLS-DA $p = 2.2 \times 10^{-22}$
CVAE $p = 1.1 \times 10^{-23}$
0.51
0.87 0.87
0.87 0.87
age
PLS-DA
CVAE
model alone
model + age
Predicted risk does not follow age
d
AD and controls age-matched (median 86 vs 85 years)
risk vs age within group: controls ρ = +0.17 (p = 0.089), AD ρ = -0.20 (p = 0.053)
predicted P(AD), PLS-DA
Control
AD
age at death (years)
Inside age bands
e
years; AD / control donors
ROC-AUC
PLS-DA
CVAE
age only
<80
21/21
80-89
49/49
90+
27/28